From domino tilings to generating functions

Indecomposable blocks, symmetric words, and Burnside's lemma

1. What are we counting?

For every integer $n \geq 1$, tile a $3 \times 2n$ board with unlabelled 1×2 dominoes. Both orientations are allowed, with no gaps, overlaps, or additional tatami restriction. A domino is an unordered pair of edge-adjacent cells. Let $\mathcal{T}_n$ be the set of all tilings. Tilings related by top-bottom reflection, left-right reflection, or a half-turn count as one type. We seek the number a_n of types, and set $a_0 = 1$ for the unique empty tiling.

$$h(r, c) = (4 - r, c), \qquad v(r, c) = (r, 2n + 1 - c), \qquad \rho = hv.$$

Since $3 \neq 2n$, the board is not square. Its full symmetry group is $G = \{e, h, v, \rho\} \cong C_2 \times C_2$; quarter-turns are not included.

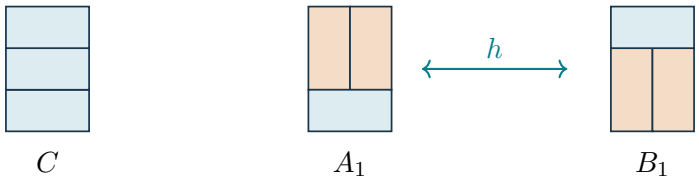


Figure 1. The three tilings of a 3×2 board. The tiling C forms one type; A_1 and B_1 form the other, so $a_1 = 2$. Colors indicate orientation only and are not labels.

The proof route

1. Use Burnside's lemma to reduce orbit counting to four fixed-point counts (p. 2).
2. Classify blocks through cut states and prove unique decomposition into a word (pp. 3–4).
3. Translate geometric symmetries into letter exchange and reversal, then construct generating functions (pp. 5–8).
4. Combine the four terms to obtain the formula, initial values, and recurrence (pp. 9–10).

Scope: these notes explain the three-row block proof, not the half-board proof or odd-height extensions. They are an exposition, not a new independent review or formal verification.

2. Burnside: why add four counts and divide by four?

The orbit $G \cdot T = \{gT : g \in G\}$ is one tiling type. The stabilizer $\mathrm{Stab}(T) = \{g : gT = T\}$ records the transformations fixing that tiling. Different transformations may produce the same tiling, so dividing the total number of tilings by four is not generally correct.

Orbit size and stabilizer size compensate

Fix T and let $K = \mathrm{Stab}(T)$. The map $g \mapsto gT$ is onto $G \cdot T$. If $U = g_0 T$, then

$$gT = U \iff g_0^{-1} g \in K \iff g \in g_0 K.$$

Thus each target tiling has a coset as its preimage, of size $|K|$. These preimages partition G, giving

$$|G \cdot T| \, |\mathrm{Stab}(T)| = |G| = 4.$$

Count the same set in two ways

$$\Omega = \{(g, T) \in G \times \mathcal{T}_n : gT = T\}.$$

Counting first by the transformation and then by the tiling gives

$$\sum_{g \in G} |\,\mathrm{Fix}(g)| = |\Omega| = \sum_{T \in \mathcal{T}_n} |\,\mathrm{Stab}(T)|.$$

Group the right-hand sum by orbits. An orbit O has $|O|$ tilings, each with stabilizer size $4/|O|$. Its total contribution is exactly 4.

Orbit size	Stabilizer size	Contribution
4	1	$4 \cdot 1 = 4$
2	2	$2 \cdot 2 = 4$
1	4	$1 \cdot 4 = 4$

Write $d_n = |\mathcal{T}_n|$, $h_n = |\,\mathrm{Fix}(h)|$, $v_n = |\,\mathrm{Fix}(v)|$, and $r_n = |\,\mathrm{Fix}(\rho)|$. The identity fixes every tiling, hence

$$\boxed{a_n = \frac{d_n + h_n + v_n + r_n}{4}.}$$

The fixed-point sets may overlap. Extra appearances of symmetric tilings compensate for their smaller orbits. All four abstract group elements fix the empty tiling, so $a_0 = (1+1+1+1)/4 = 1$; faithfulness is not required.

3. Turn a tiling into a state path

For an arbitrary width w, let $S_j \subseteq \{1,2,3\}$ record the rows of horizontal dominoes crossing the cut between columns j and $j+1$. At the board boundaries, $S_0 = S_w = \varnothing$. An internal cut is a fault, where the tiling can be separated, exactly when $S_j = \varnothing$.

The local rule for one column

Let p, q be the incoming and outgoing states of a column. Cells in p are covered from the left; cells in q extend to the right. These sets must be disjoint. The remaining cells must be paired vertically within the column, so their row set must be empty or a pair of adjacent rows.

$$\boxed{p \to q \iff p \cap q = \varnothing, \quad \{1,2,3\} \setminus (p \cup q) \in \{\varnothing, \{1,2\}, \{2,3\}\}.}$$

Input p	All legal outputs q
$\varnothing$	$\{1\}, \{3\}, \{1,2,3\}$
$\{1\}$	$\varnothing, \{2,3\}$
$\{2\}$	$\{1,3\}$
$\{3\}$	$\varnothing, \{1,2\}$
$\{1,2\}$	$\{3\}$
$\{1,3\}$	$\{2\}$
$\{2,3\}$	$\{1\}$
$\{1,2,3\}$	$\varnothing$

For example, if $p = \{3\}$, the bottom cell is already covered. The top two cells either form one vertical domino, giving empty output, or start two horizontal dominoes, giving output $\{1,2\}$.

Why is this a bijection, not just a necessary condition?

A tiling gives a unique legal path. Conversely, place a crossing horizontal domino in every row specified by each S_j, then add the unique vertical completion in every column. Disjointness prevents overlap, the complement rule guarantees coverage, and empty endpoints prevent protrusion. These constructions are inverse.

$$\boxed{\text{Complete tilings} \longleftrightarrow \text{Legal paths from the empty state to itself.}}$$

The states $\{2\}$ and $\{1,3\}$ only lead to each other and are unreachable from the empty state. They are still included in the complete table. One transition processes one column, not one block.

4. Classify the blocks and prove unique decomposition

A block is a nonempty complete tiling with no internal fault. It corresponds to a first-return path from the empty state. There are only three possible first steps.

- C: $\varnothing \to \{1,2,3\} \to \varnothing$. The second step is forced; this gives three horizontal dominoes.
- A_k: Start at $\{3\}$, repeat $\{3\} \to \{1,2\} \to \{3\}$ exactly $k-1$ times, then return to the empty state. The length is $2k$.
- B_k: Start at $\{1\}$ and alternate through $\{2,3\}$ in the same way. This is the top-bottom reflection of A_k.

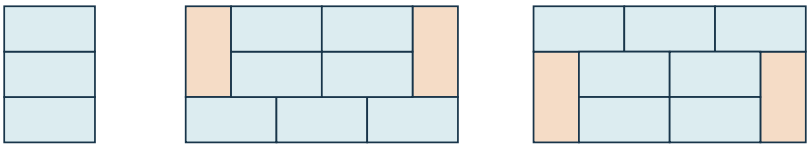


Figure 2. Representatives of the three block types. In A_k, endpoint vertical dominoes cover rows 1 and 2. Bottom-row dominoes use column pairs $(2j-1, 2j)$; internal dominoes in the top two rows use $(2j, 2j+1)$. Thus odd internal cuts are crossed in the bottom row and even cuts in the top two rows. Every internal cut is nonempty. The transition table allows no other first step or nonterminal continuation. This is an exhaustive classification, not an extrapolation from pictures. When $k = 1$, no loop is taken; C, A_1, B_1 are distinct width-two blocks.

$$\mathcal{B} = \{C\} \cup \{A_k, B_k : k \geq 1\}.$$

Cut at all faults, and only at faults

Cut a complete tiling at every fault. Each piece is a block. Every block boundary must be a fault, and omitting a fault leaves a decomposable piece. Thus the decomposition is unique. Conversely, concatenate any finite word of blocks; cutting it again recovers precisely that word.

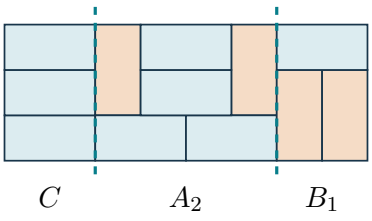


Figure 3. The unique boundaries of CA_2B_1 are dashed. There are three blocks, but the half-width is $1+2+1=4$. The empty tiling corresponds to the empty word.

5. How do geometric symmetries act on words?

Every block is individually left-right symmetric

For A_k, reflect columns by $c \mapsto 2k+1-c$. The endpoint vertical dominoes are exchanged. Bottom-row pairs $(2j-1, 2j)$ map to pairs of the same form with index $k+1-j$; top-row pairs $(2j, 2j+1)$ map to pairs with index $k-j$. The set of dominoes is preserved. The block C is fixed too. Since the reflections commute, $B_k = h(A_k)$ is also left-right symmetric. Define the weight-preserving involution $\sigma(C) = C$, $\sigma(A_k) = B_k$, $\sigma(B_k) = A_k$. Symmetries preserve faults. Unique decomposition therefore gives

$$\boxed{h(W) = \sigma(W), \qquad v(W) = \mathrm{rev}(W), \qquad \rho(W) = \sigma(\mathrm{rev}(W)).}$$

Here σ acts letter by letter, while rev reverses the order. Block widths need not agree: the faults move with the tiling.

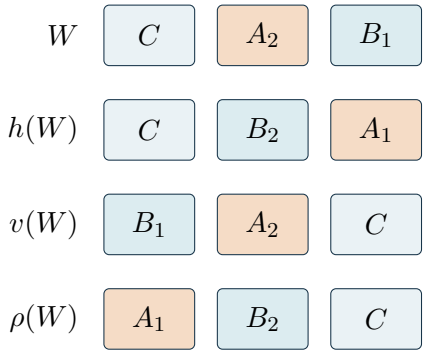


Figure 4. The three operations on a word. Boxes represent letters, not their geometric widths.

Three fixed-word conditions

- $h(W) = W$: Every letter is fixed by σ, so only C can occur.
- $v(W) = W$: $b_i = b_{\ell+1-i}$. Opposite letters agree: an ordinary palindrome.
- $\rho(W) = W$: $b_i = \sigma(b_{\ell+1-i})$. Opposite letters are paired as A_k, B_k or C, C.

Unique decomposition lets us infer equality of words from equality of tilings. For example, A_2B_2 is not top-bottom symmetric: reflection gives B_2A_2 without exchanging the left and right positions.

6. Generating functions: from one block to any word

The exponent of x records half-width, not the number of blocks. Set $\mathrm{wt}(C) = 1$ and $\mathrm{wt}(A_k) = \mathrm{wt}(B_k) = k$; word weights add. Write $[x^n]S(x)$ for the coefficient of x^n.

Why define F this way?

Assign a monomial $x^{\mathrm{wt}(b)}$ to each block and add them. Blocks of the same half-width contribute the same power, so its coefficient counts them. There are three width-two blocks and two blocks at each larger even width.

$$F(x) = \sum_{b\in\mathcal{B}} x^{\mathrm{wt}(b)} = x + 2\sum_{k\geq 1} x^k = 3x + 2x^2 + \cdots = \frac{3x - x^2}{1 - x}.$$

Independent choices multiply; concatenated sizes add

Choosing ℓ blocks in order means selecting one term from each of ℓ copies of $F(x)$. Multiplication gives

$$F(x)^\ell = \sum_{(b_1,\ldots,b_\ell)\in\mathcal{B}^\ell} x^{\mathrm{wt}(b_1)+\cdots+\mathrm{wt}(b_\ell)}.$$

For example, CA_2 and A_2C each contribute x^3. Their contributions add; commutativity does not identify the two choices. Symmetry classes are not being counted yet. The classes distinguished by block count are disjoint because decomposition is unique. Hence

$$\boxed{D(x) = \sum_{\ell\geq 0} F(x)^\ell = \frac{1}{1 - F(x)} = \frac{1 - x}{1 - 4x + x^2}.}$$

The constant 1 represents the empty word. Alternatively, a word is empty or consists of one first block followed by an arbitrary word, giving $D = 1 + FD$.

Example: total half-width three

Blocks	Ordered half-widths	Count
1	(3)	2
2	$(1,2),(2,1)$	$3\cdot 2 + 2\cdot 3 = 12$
3	$(1,1,1)$	$3^3 = 27$

Thus $d_3 = [x^3]D = 2 + 12 + 27 = 41$.

No numerical convergence is needed. Since $F(0) = 0$, the lowest degree of F^ℓ is at least ℓ, so only $\ell \leq n$ contribute to $[x^n]$. The finite identity $(1 - F)\sum_{\ell=0}^{L} F^\ell = 1 - F^{L+1}$ proves the infinite identity coefficient by coefficient.

7. Left-right symmetry: why does $D(x^2)$ appear?

A palindrome has one of two unique forms, according to the parity of its number of blocks:

$$U\,\mathrm{rev}(U), \qquad Ub\,\mathrm{rev}(U) \quad (b \in \mathcal{B}).$$

The left word U is arbitrary, but the right word is forced to be its reversal. If U has weight m, the two sides have weight $2m$, with only d_m choices, not two independent choices. Their generating function is

$$\sum_{m\geq 0} d_m x^{2m} = D(x^2).$$

The center contributes 1 if empty and $F(x)$ if it contains one arbitrary block. This choice is independent of U, hence

$$\boxed{V(x) = (1 + F(x))D(x^2).}$$

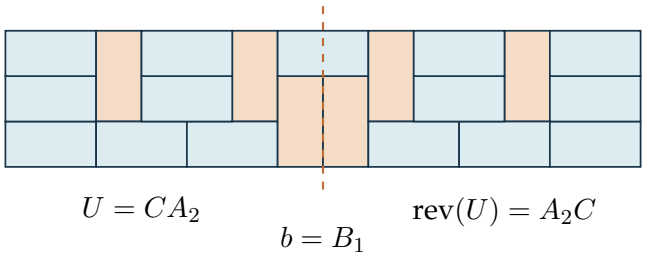


Figure 5. The palindrome $CA_2B_1A_2C$. The axis passes through the central block B_1, not a block boundary; dominoes may cross the geometric midline.

Why not $D(x)^2$?

The product $D(x)^2$ counts ordered pairs of independent words. Its coefficients are

$$[x^n]D(x)^2 = \sum_{i+j=n} d_i d_j.$$

It imposes neither equal widths nor reversal. Even two weight-one words give nine choices, while palindromes allow only CC, A_1A_1, B_1B_1.

Example: a 3×6 board

The total weight is 3, so a center block is necessary. With a center of weight k and a left word of weight m, we need $2m + k = 3$:

- $(m, k) = (0, 3)$: The central block is A_3 or B_3: two choices.
- $(m, k) = (1, 1)$: U and the central block each have three choices: nine total.

Thus $v_3 = 2 + 9 = 11$. In general the expression simplifies to

$$V(x) = \frac{1 + 2x - x^2}{1 - x}\frac{1 - x^2}{1 - 4x^2 + x^4} = \frac{1 + 3x + x^2 - x^3}{1 - 4x^2 + x^4}.$$

8. Top-bottom symmetry and half-turn symmetry

Top-bottom symmetry: every position must be fixed

Top-bottom reflection preserves the order and exchanges A_k, B_k. Thus $h(W) = W$ requires every letter to be fixed by σ, so only C is allowed. The words are $\varepsilon, C, C^2, \ldots$. Since each C has weight one,

$$\boxed{H(x) = \sum_{n\geq 0} x^n = \frac{1}{1-x}, \qquad h_n = 1.}$$

Half-turn symmetry: pair the two sides using σ

A half-turn-fixed word satisfies $b_i = \sigma(b_{\ell+1-i})$. Once the first half U is chosen, the second half must be $\sigma(\mathrm{rev}(U))$. A central block, if present, must satisfy $b = \sigma(b)$, so it can only be C. The two forms are

$$U\sigma(\mathrm{rev}(U)), \qquad UC\sigma(\mathrm{rev}(U)).$$

The involution σ preserves weight, so the two sides still contribute $D(x^2)$. The center is empty or one C, contributing $1 + x$. Thus

$$\boxed{R(x) = (1+x)D(x^2) = \frac{1 + x - x^2 - x^3}{1 - 4x^2 + x^4}.}$$

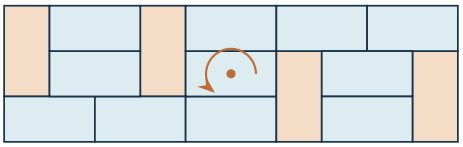


Figure 6. The word A_2CB_2 is fixed by a half-turn, but by neither reflection separately. The end blocks exchange both position and type. The coefficients also give the useful identities

$$R(x) = \sum_{m\geq 0} d_m x^{2m} + \sum_{m\geq 0} d_m x^{2m+1}, \qquad \boxed{r_{2m} = r_{2m+1} = d_m.}$$

For total half-width three, U has weight one and the center is C, giving exactly CCC, A_1CB_1, B_1CA_1. The left-right symmetric count is eleven because its center may be any block.

The presence of a center block is determined by the parity of the block count, not initially by half-width. For a half-turn, the center can only be the weight-one block C, so these parities then correspond. Equal widths of paired blocks align the word center with the geometric center.

9. Combine the four generating functions

Multiply the Burnside identity for each n by x^n and sum:

$$\boxed{A(x) = \sum_{n\geq 0} a_n x^n = \frac{D(x) + H(x) + V(x) + R(x)}{4}.}$$

Write $P = 1 - 4x + x^2$ and $Q = 1 - 4x^2 + x^4$. We have established

$$D(x) = \frac{1-x}{P}, \qquad H(x) = \frac{1}{1-x},$$
$$V(x) = \frac{1 + 3x + x^2 - x^3}{Q}, \quad R(x) = \frac{1 + x - x^2 - x^3}{Q}.$$

First combine the two terms with denominator Q:

$$V(x) + R(x) = \frac{2 + 4x - 2x^3}{Q}.$$

Hence

$$A(x) = \frac{1}{4}\left[\frac{1-x}{P} + \frac{1}{1-x} + \frac{2(1 + 2x - x^3)}{Q}\right].$$

Putting the terms over a common denominator

The common denominator is $(1-x)PQ$, and its numerator is

$$\frac{1}{4}\big((1-x)^2 Q + PQ + 2(1 + 2x - x^3)(1-x)P\big).$$

For a direct check, the three polynomial terms expand as follows:

$$\begin{aligned}(1-x)^2 Q &= 1 - 2x - 3x^2 + 8x^3 - 3x^4 - 2x^5 + x^6,\\ PQ &= 1 - 4x - 3x^2 + 16x^3 - 3x^4 - 4x^5 + x^6,\\ 2(1 + 2x - x^3)(1-x)P &= 2 - 6x - 10x^2 + 16x^3 + 6x^4 - 10x^5 + 2x^6.\end{aligned}$$

Adding and dividing by four gives $1 - 3x - 4x^2 + 10x^3 - 4x^5 + x^6$. Therefore

$$\boxed{A(x) = \frac{1 - 3x - 4x^2 + 10x^3 - 4x^5 + x^6}{(1-x)(1 - 4x + x^2)(1 - 4x^2 + x^4)}.}$$

This is a coefficient identity for all $n \geq 0$. Neither the target formula nor a list of initial values was used to guess the block classification. The classification and bijections came first; the generating functions follow from them.

Negative coefficients in the numerator are not negative tiling counts. The numerator alone is not a counting series; the count is $[x^n]A(x)$ after expanding the rational function as a formal power series.

10. Initial values, recurrence, and the completed argument

Comparing coefficients in $(1-4x+x^2)D(x) = 1-x$ gives $d_0 = 1, d_1 = 3$ and $d_n = 4d_{n-1} - d_{n-2}$ for $n \geq 2$. Expand H, V, R and apply Burnside coefficientwise:

n	d_n	h_n	v_n	r_n	a_n
0	1	1	1	1	1
1	3	1	3	1	2
2	11	1	5	3	5
3	41	1	11	3	14
4	153	1	19	11	46
5	571	1	41	11	156
6	2131	1	71	41	561

For example, $a_4 = (153 + 1 + 19 + 11)/4 = 46$. The denominator expands to

$$1 - 5x + x^2 + 19x^3 - 19x^4 - x^5 + 5x^6 - x^7.$$

The numerator has degree six, so for $n \geq 7$ the coefficient of x^n in the product of this denominator and $A(x)$ is zero. Rearranging gives

$$\boxed{\begin{aligned} a_n &= 5a_{n-1} - a_{n-2} - 19a_{n-3} + 19a_{n-4} \\ &\quad + a_{n-5} - 5a_{n-6} + a_{n-7}, \qquad n \geq 7. \end{aligned}}$$

Logical dependencies and boundary cases

The complete local rule gives the tiling-path bijection. First returns exhaust the blocks. Unique faults prevent different words from representing the same tiling. The symmetry actions and fixed-word classifications give the four fixed-point counts. Burnside converts them into orbit counts. These arguments cover arbitrary lengths; figures and numerical tables illustrate and check them. The empty word is included in all four series; $k = 1$ allows zero loops; unequal block widths cause no problem for reflection; the geometric axis may cross a central block; all infinite sums are well defined coefficientwise in $\mathbb{Q}[[x]]$.

Source and status of these notes

Primary source: Shiqi Liu, *Domino Tilings of Odd-Height Rectangles up to Symmetry*, manuscript dated October 25, 2026, Sections 1–2 (PDF pp. 1–5), together with the explanations developed in our discussion. These notes reorganize that proof route and add no new research claim. Bibliographic context, following the manuscript: the formula corresponds to Table 10, Conjecture 6 of Mathar's *Paving rectangular regions with rectangular tiles: tatami and non-tatami tilings* (arXiv:1311.6135), and OEIS A232622. External status and priority were not rechecked for this exposition; no priority claim is made. Production note: AI-assisted exposition and bilingual typesetting; diagrams are generated from the defining coordinates. Layout, transcription, and finite algebra checks are not external mathematical review or a claim of complete Lean verification of this route.